Data Safari #1

Week 4 · Replicating a published figure

This is the first of four Data Safari exercises this semester (Weeks 4, 7, 10, and 13). See the Assignments overview for the general format — the instructions below are the same each time, applied to a new figure of your choosing. Render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas alongside this week's regular assignment, before Week 5's lecture (Monday). See the syllabus for the late policy.

Instructions

1. **Find a paper** in your own subfield with (a) a figure showing real data — a scatterplot, boxplot, or bar chart is easiest to start with — and (b) publicly available data, either in a supplement, a data repository (Dryad, Figshare, GenBank, etc.), or linked in the paper's data-availability statement.

Full citation:

2. **Recreate the figure** as closely as you can using R and `ggplot2`, working from the actual data file.

3. **Reflect**, in a few sentences each:
   - How easy was it to find and load the data? Were column names, units, and structure clear from the paper and/or a data dictionary?
   - How closely does your version match the published figure? Note any differences in scale, color, or labeling, and your best guess at why they exist.
   - If you were a reviewer, what one change to the paper's methods or data documentation would have made this exercise easier?